

HINTS & SOLUTIONS

Q.3 (C)

The initial gravitational potential energy of the chain

$$= U_1 = m_1 g h_1$$

$$\Rightarrow U_1 = Mg \frac{\ell}{2}$$

The final gravitational potential energy of the chain = $U_2 = m_2 g h_2$

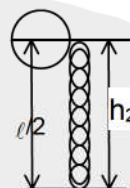
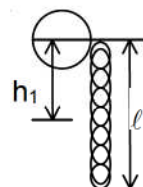
$$U_2 = \left(\frac{M}{2}\right) g \frac{\ell}{4} = \frac{Mg\ell}{8}$$

$\therefore$ The work done by the external agent =

$$W = |U_1 - U_2|$$

$$= Mg \frac{\ell}{2} \left[1 - \frac{1}{4}\right]$$

$$= \frac{3}{8} Mg\ell$$



Q.5 (A)

Let the spring be elongated by x . That means the pulley falls through a distance x . Consequently the string in both sides of the pulley is slackened to the same extent x . Therefore the total distance moved by the body = $2x$.

$\Rightarrow$ Since the spring is elongated by x ,

$$(\Delta PE)_{\text{spring}} = \frac{1}{2} kx^2$$

As the body falls through a distance $2x$.

$$(\Delta PE)_{\text{gr}} = -mg(2x).$$

Since the body is initially at rest and comes to rest just at the instant of maximum elongation of the spring

$$\Rightarrow \Delta PE = 0$$

Applying the principle of conservation of energy,

$$(\Delta PE) + (\Delta KE) = 0$$

$$(\Delta PE)_{\text{spring}} + (\Delta PE)_{\text{gr}} + \Delta KE = 0$$

$$\Rightarrow \frac{1}{2} kx^2 - mg(2x) + 0 = 0$$

$$\Rightarrow x = \frac{4mg}{k}$$

Q.14 (D)

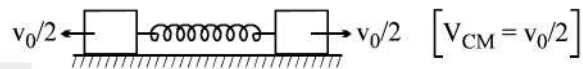
$$v_2 = 2u + v_1$$

$$v_2 = 2 \times 5 + 10 = 20 \text{ m/s}$$

$$\vec{J} = \Delta \vec{p} = \vec{p}_f - \vec{p}_i = 1 \times 20 - (-1 \times 10) = 30 \text{ N-s}$$

Q.16 (ABD)

In CM frame the blocks will perform SHM as



$\therefore V_{CM}$ is constant hence both will have maximum velocity $v_0/2$ towards right in CM frame.

$$\therefore (v_B)_{\max} = v_0 \quad \text{in ground frame}$$

$$\text{In CM frame } (v_A)_{\min} = -v_0/2$$

$$\therefore (v_A)_{\min} = 0 \quad \text{in ground frame}$$

i.e. A move only in right direction

In CM frame $v_B = v_0/2$ left in the case when spring is at natural length

$$\therefore v_B = 0 \quad \text{in ground frame at that instant.}$$

Q.17 (AB)

At maximum height, relative velocity between block and wedge is zero. So, their common velocity is also the velocity of centre of mass. So long as the block is on the wedge, one component of normal reaction keeps on accelerating the wedge.

Q.18 (ACD)

Some heat is also dissipated

Q.19 (ABC)

$$(A) F = \frac{mdv}{dc} = 10 \times \frac{15-0}{3} = 50 \text{ N}$$

$$(B) 10 \times 15 + 25v = 35 \times 5$$

$$25v = 175 - 150$$

$$v = 1 \text{ m/s}$$

Q.20 (AC)

$$3 \times 2\hat{i} + 2 \times 1\hat{j} = 3 \times (1 \cos 30^\circ \hat{i} + 1 \sin 30^\circ \hat{j}) + 2 \times \vec{v}$$

$$\left(36 - \frac{3\sqrt{3}}{4}\right)\hat{i} + \frac{1}{4}\hat{j} = \vec{v}$$

$$E_i = \frac{1}{2} \times 3 \times 2^2 + \frac{1}{2} \times 2 \times 1^2 = 7 \text{ J}$$

$$E_s = \frac{1}{2} \times 3 \times 1^2 + \frac{1}{2} \times 2 \times \left[\left(\frac{1}{4}\right)^2 + \left(3 - \frac{3\sqrt{3}}{4}\right)^2 \right]$$

$$= 1.5 + 1 \left(\frac{1}{16} + \frac{9(16+3-8\sqrt{3})}{16} \right)$$

$$= \frac{3}{2} + \frac{172 - 72\sqrt{3}}{16} = 1.5 + 10.75 - 4.5\sqrt{3} = 12.25 - 4.5\sqrt{3}$$



Q.22 $\left[\frac{(2m_1 + m_2)g}{k} \right]$

In order to lift the block m_2 from the surface, the spring force F must be equal to the weight of the block m_2 , at position 2, (say). Let the corresponding elongation of the spring be x_2 .

Hence

$$kx_2 \Rightarrow m_2 g \Rightarrow x_2 = m_2 g/k \quad \dots(1)$$

$$\Rightarrow \text{P.E. stored in the spring in position 2} = \frac{1}{2} kx_2^2$$

Let, initially the spring be compressed by x_1 .

$$\text{Therefore, the potential energy stored in the spring at this position} = \frac{1}{2} kx_1^2$$

$$(\Delta PE)_{\text{spring}} = \frac{1}{2} kx_2^2 - \frac{1}{2} kx_1^2$$

$$= \frac{1}{2} k(x_2^2 - x_1^2) \quad \dots(2)$$

As the block rises through a distance $(x_1 + x_2)$, change in its gravitational potential energy is,

$$= (\Delta PE)_{\text{gr}} = mg(x_1 + x_2) \quad \dots(3)$$

Since the blocks are at rest at both the position 1 and 2

$$(\Delta KE)_{\text{system}} = 0$$

Conserving the energy of the system between position 1 and 2

$$\Delta PE + \Delta KE = 0$$

$$(\Delta PE)_{\text{gr}} + (\Delta PE)_{\text{spring}} = 0$$

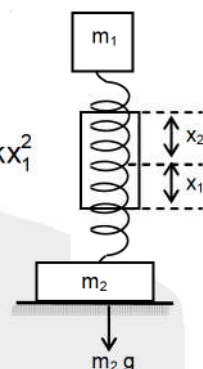
$$m_1 g(x_1 + x_2) + \frac{1}{2} k(x_2^2 - x_1^2) = 0$$

$$\Rightarrow m_1 g + \{k(x_2 - x_1)\}/2 = 0$$

$$\text{putting } x_2 = \frac{m_2 g}{k} \text{ we obtain,}$$

$$m_1 g + \frac{k}{2} \left(\frac{m_2 g}{k} - x_1 \right) = 0 \Rightarrow (2m_1 + m_2) g = kx_1$$

$$\Rightarrow x_1 = \frac{(2m_1 + m_2)g}{k}$$



Q.23 $\pi \rho v^2 d^2 / 4$

Q.24 $\cos^2 \theta$